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Project Proposal: Face Mask Detection

Why is this important?

Why are face masks important in 2026?

Problem:

Not many people enjoy wearing face masks, but they are required for things, so people will sometimes just not wear them or wear them wrong.

Who cares?

Mostly people in the medical field.

Why is it important?

It helps prevent the spread of disease in many medical environments.

Overview of My Solution:

Face Mask Detection:

What does it do?

This will identify if it is both on their face, and on it correctly.

How is it solving?

Object detection will identify the mask, as well as if it is covering the nose.

A TL:DR.

Image taken -> Faster R-CNN detects mask/nose -> identifies if mask is present and nose is not -> Generates results

Technical Specs

CV Technique:

Object detection

Model:

Faster R-CNN

Framework:

Pytorch

Justifications:

Previous experience in this course, great object detection.

Data Plan

It's gotta come from somewhere!

Source:

Kaggle image set.

Size:

About 1,000 is the goal.

Labels:

Mask presence, mask on correctly.

Preparation:

Not Needed (image set)

System Diagram

Input:
Image,
hopefully with a
mask.

Faster R-CNN
identifies
objects

Faster R-CNN
Identifies if
mask is on and
covering
mouth/nose

Report
generated of
results

Output: Report
given to user

Success Metrics

Primary Metric: Accuracy
Goal is >90%

Secondary Metric: Precision
Goal is >90%

Timeline

Week 10

Get Dataset, set up environment.

Goal: Dataset Ready

Week 11

Train or fine-tune model

Goal: Model Working

Week 12

Test and Improve

Goal: Hit accuracy and precision goals

Week 13

Create demo and video

Goal: Demo ready

Week 14

Final testing, documentation

Goal: Everything Done

Plan B's

Nothing goes perfectly!

Challenge: Not enough data

Solution: Use ImageNet set

Challenge: Low accuracy

Solution: fine tune, and if that doesn't work, find more data.

Resources Needed

No money!

Compute: Colab

Framework: PyTorch